QUANTUM SECURITY EXPERT ROADMAP

Zero to Enterprise-Deliverable // Self-Study Track

Compiled for: Ribbz (Jacob Ribbe)

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ROADMAP OVERVIEW

8 phases. ~12 months to expert-level delivery capability. Each phase builds on the last. Every phase has a concrete deliverable that proves competence and builds portfolio. The goal is not academic knowledge -- it's the ability to walk into a Fortune 500 security team and deliver a solution they'd pay for.

Phase	Domain	Weeks	Outcome
1	Math Foundations	2-3	Crypto math fluency
2	Classical Cryptography	3-4	Working implementations
3	Quantum Mechanics for CS	3-4	Qiskit simulations
4	Quantum Threat Model	2-3	Shor/Grover attack demos
5	PQC Standards	3-4	Lattice crypto implementations
6	Build: Migration Scanner	4-5	Shippable tool v1
7	Quantum Networking + QKD	3-4	QKD protocol demos
8	Certs + Portfolio + GTM	4-6	Enterprise-ready profile

Total: ~28-35 weeks core + ongoing cert prep. Your learning velocity (top 3%) compresses this. Realistic target: expert-level in 10-12 months with consistent focus.

PHASE 1 // MATH FOUNDATIONS

Weeks 1-3 // Prerequisite layer

You cannot fake the math. Every phase after this depends on these fundamentals. The good news: your ML coursework already gave you some linear algebra exposure. This phase fills the gaps specific to cryptography.

Core Topics

- Modular arithmetic — the backbone of all public-key crypto (mod, GCD, Euler's totient)
- Group theory basics — cyclic groups, generators, order. RSA and ECC live here
- Finite fields (GF(p), GF(2^n)) — AES operates in GF(2^8), ECC uses prime fields
- Linear algebra refresh — vector spaces, lattices, matrix operations (PQC foundation)
- Probability + information theory — entropy, randomness, birthday paradox (hash collisions)
- Number theory — primes, factorization, discrete logarithm problem (what Shor breaks)

Resources

>> "An Introduction to Mathematical Cryptography" — Hoffstein, Pipher, Silverman (THE textbook)

>> 3Blue1Brown — Essence of Linear Algebra (YouTube, free)

>> Khan Academy — Number Theory + Modular Arithmetic modules

>> MIT OCW 6.042J — Mathematics for Computer Science (free)

Phase 1 Deliverable

>> Python library: modular_math.py — implement modular exponentiation, extended Euclidean algorithm, Miller-Rabin primality test, basic finite field operations. You'll reuse this in Phase 2.

PHASE 2 // CLASSICAL CRYPTOGRAPHY

Weeks 4-7 // Understand what you're protecting

You need to deeply understand the systems quantum will break before you can defend against the break. Build each algorithm from scratch in Python.

Core Topics

- Symmetric crypto — AES (build it), block cipher modes (CBC, GCM, CTR), stream ciphers
- Asymmetric crypto — RSA (build it from your Phase 1 math), Diffie-Hellman key exchange
- Elliptic Curve Cryptography — point addition, scalar multiplication, ECDSA, ECDH
- Hash functions — SHA-256 internals, HMAC, Merkle trees, collision resistance
- Digital signatures — RSA-PSS, ECDSA, Ed25519
- TLS/SSL protocol — how all the above combines in real-world HTTPS
- Key management — generation, storage, rotation, PKI, certificate chains

Resources

>> "Serious Cryptography" — Jean-Philippe Aumasson (practical, modern, no fluff)

>> "Crypto101" — free online textbook (crypto101.io)

>> CryptoPals challenges — cryptopals.com (learn by breaking, 48 challenges)

>> "Understanding Cryptography" — Christof Paar (textbook + YouTube lectures)

Phase 2 Deliverables

>> 1. Working Python implementations of RSA, AES, ECDH, SHA-256 (from scratch, not libraries)

>> 2. Complete at least Sets 1-4 of CryptoPals (proves you can break crypto, not just build it)

>> 3. TLS handshake analyzer script — capture and decode a real HTTPS handshake

PHASE 3 // QUANTUM MECHANICS FOR COMPUTER SCIENTISTS

Weeks 8-11 // Just enough physics, no more

You don't need a physics degree. You need to understand qubits, gates, and measurement well enough to reason about quantum algorithms. This is linear algebra applied to a specific physical system. Your hyperphantasia is a massive advantage here — quantum states are geometric objects in vector spaces.

Core Topics

- Qubit representation — Bloch sphere, Dirac notation, state vectors
- Superposition + entanglement — what they actually mean mathematically (tensor products)
- Quantum gates — Hadamard, CNOT, Pauli-X/Y/Z, Toffoli, phase gates
- Quantum circuits — composition, measurement, collapse
- Quantum Fourier Transform — the foundation of Shor's algorithm
- Qiskit fundamentals — build and run circuits on IBM simulators + real hardware
- Noise models — decoherence, gate errors, why error correction matters

Resources

>> "Quantum Computing: An Applied Approach" — Jack Hidary (best for CS people, not physicists)

>> IBM Qiskit Textbook — free, interactive, runs on real quantum hardware

>> "Quantum Computation and Quantum Information" — Nielsen & Chuang (the bible, use as reference)

>> Microsoft Quantum Katas — learn-by-coding quantum exercises

Phase 3 Deliverables

>> 1. Qiskit notebook: build + run 10 fundamental quantum circuits on real IBM hardware

>> 2. Implement quantum teleportation protocol from scratch

>> 3. QFT implementation with visualization of state evolution

PHASE 4 // QUANTUM THREAT MODEL

Weeks 12-14 // Understand the attack surface

This is where your Phase 2 (classical crypto) and Phase 3 (quantum) knowledge converge. You'll understand exactly WHY quantum breaks RSA and ECC, simulate the attacks, and quantify the timeline. This is the knowledge that makes you dangerous in a boardroom.

Core Topics

- Shor's algorithm — full implementation on a simulator, understand why it factors integers in polynomial time
- Grover's algorithm — quadratic speedup on search, implications for symmetric crypto (AES-128 -> AES-256)
- Harvest Now, Decrypt Later (HNDL) — the real-world threat model driving enterprise urgency
- Quantum threat timeline — NIST estimates, Q-Day projections, what's realistic vs hype
- Mosca's inequality — framework for when organizations need to start migrating
- Impact mapping — which algorithms break (RSA, ECC, DH) vs which survive (AES-256, SHA-256)

Resources

>> NIST Post-Quantum Cryptography project page — csrc.nist.gov/pqc

>> Qiskit implementation of Shor's algorithm — IBM tutorial

>> "Quantum Computing Since Democritus" — Scott Aaronson (conceptual depth)

>> ETSI Quantum Safe Cryptography reports — industry threat assessments

Phase 4 Deliverables

>> 1. Shor's algorithm simulation: factor a small semiprime on Qiskit + explain each step

>> 2. Grover's algorithm: demonstrate search speedup + calculate AES key-space implications

>> 3. Quantum Threat Assessment template — a document you could present to a CISO

PHASE 5 // POST-QUANTUM CRYPTOGRAPHY STANDARDS

Weeks 15-18 // The defense layer

NIST finalized PQC standards in 2024. This is the most commercially valuable phase. Companies are actively migrating and need people who understand these algorithms deeply enough to implement, audit, and troubleshoot them. The math here connects directly to your ML knowledge — lattices, vector spaces, hardness assumptions.

Core Topics

- Lattice-based cryptography — Learning With Errors (LWE), Ring-LWE, shortest vector problem
- ML-KEM (formerly CRYSTALS-Kyber) — NIST's primary key encapsulation mechanism
- ML-DSA (formerly CRYSTALS-Dilithium) — NIST's primary digital signature algorithm
- SLH-DSA (formerly SPHINCS+) — hash-based signatures, stateless
- FN-DSA (formerly FALCON) — compact lattice-based signatures
- Hybrid schemes — running classical + PQC in parallel during migration
- Performance analysis — key sizes, computation overhead, bandwidth implications
- PQC in TLS 1.3 — how Chrome/Firefox are already deploying hybrid PQC

Resources

>> NIST FIPS 203, 204, 205 — the actual standards (read the source)

>> Open Quantum Safe (OQS) project — liboqs library, reference implementations

>> "Post-Quantum Cryptography" — Bernstein, Buchmann, Dahmen (advanced textbook)

>> PQShield + Cloudflare PQC blog posts — real-world deployment insights

Phase 5 Deliverables

>> 1. Python implementation of simplified LWE encryption (from math, not a library)

>> 2. Benchmark suite: ML-KEM vs RSA vs ECDH — key generation, encapsulation, decapsulation times

>> 3. PQC migration guide document — protocol-level recommendations for TLS, SSH, VPN, email

PHASE 6 // BUILD: PQC MIGRATION SCANNER

Weeks 19-23 // The money deliverable

This is where everything converges into a shippable product. A tool that scans codebases, network configs, and certificate chains for quantum-vulnerable cryptography and generates actionable migration reports. This is what you sell or demo to enterprises.

Tool Architecture

- Static analysis engine — scan Python, Java, C/C++, Go, JS for crypto API calls
- Certificate chain scanner — X.509 certs, identify RSA/ECC keys, flag key sizes
- Network protocol analyzer — TLS versions, cipher suites, key exchange methods
- Dependency scanner — detect vulnerable crypto libraries in requirements/packages
- Risk scoring — classify findings by urgency (HNDL exposure, data sensitivity)
- Migration recommendations — map each finding to specific PQC replacement
- Report generator — PDF/HTML output suitable for CISO-level presentation

Tech Stack

- Python core — AST parsing, regex patterns, subprocess for openssl calls
- liboqs Python bindings — for PQC algorithm testing/validation
- ReportLab — PDF report generation (you already know this from Professor skill)
- Click or Typer — CLI framework
- Optional: FastAPI wrapper for enterprise deployment

Phase 6 Deliverables

>> 1. Working CLI tool: pqc-scanner — scans a codebase and outputs a risk report

>> 2. GitHub repo with docs, examples, and test suite

>> 3. Demo video / writeup showing the tool running against a real open-source project

PHASE 7 // QUANTUM NETWORKING + QKD

Weeks 24-27 // Expert-tier differentiation

This separates you from the PQC-only crowd. Most quantum security people stop at post-quantum algorithms. Understanding quantum key distribution and quantum networking makes you rare — there are maybe a few thousand people globally who can speak to both PQC and QKD at a technical level.

Core Topics

- BB84 protocol — the original QKD protocol, implement in simulation
- E91 protocol — entanglement-based QKD
- Quantum Random Number Generators (QRNG) — how and why they matter for key generation
- Quantum repeaters + quantum internet — the infrastructure layer
- QKD hardware landscape — who makes what (ID Quantique, Toshiba, QuintessenceLabs)
- QKD limitations — distance, cost, side-channel attacks, practical deployment challenges
- Hybrid QKD + PQC architectures — the realistic near-term security stack

Resources

>> Qiskit QKD tutorials — BB84 simulation

>> ID Quantique whitepapers — real-world QKD deployment case studies

>> "Quantum Networking" — Rodney Van Meter (the infrastructure textbook)

>> ETSI QKD standards — GS QKD series

Phase 7 Deliverables

>> 1. BB84 + E91 protocol simulations with eavesdropper detection

>> 2. QRNG implementation using IBM quantum hardware

>> 3. Technical whitepaper: 'Hybrid QKD + PQC Security Architecture' — publishable quality

PHASE 8 // CERTIFICATIONS + PORTFOLIO + GO-TO-MARKET

Weeks 28-35 // Become hireable / contractable

Knowledge without credentials is invisible to enterprises. This phase converts everything you've built into a profile that security teams and consulting firms take seriously.

Certification Path (priority order)

- CompTIA Security+ — baseline credibility, 2-3 weeks prep given your background
- ISC2 CC (Certified in Cybersecurity) — free, entry-level, gets you in the ISC2 ecosystem
- IBM Quantum Developer Certificate — proves you can build on real quantum hardware
- CISSP (long-term) — the enterprise gold standard, requires 5 years experience OR associate path
- Specialized: ISARA Quantum-Safe readiness training — PQC-specific credential

Portfolio Pieces (from all phases)

- GitHub: pqc-scanner tool (Phase 6) — your flagship deliverable
- GitHub: quantum algorithm implementations (Phases 3-4) — proves quantum literacy
- GitHub: classical crypto from-scratch implementations (Phase 2) — proves foundations
- Published whitepaper on hybrid QKD + PQC (Phase 7) — thought leadership
- Blog series: 'Quantum Security for Engineers' — content marketing that builds authority
- Demo reel: tool walkthrough + threat assessment presentation

Go-to-Market Options

- Consulting — quantum risk assessments for mid-market companies ($150-300/hr range)
- Product — pqc-scanner as SaaS or enterprise license
- Employment — quantum security roles at IBM, Google, Microsoft, AWS, or defense contractors
- Hybrid — employed + side consulting (most realistic starting point)

STRATEGIC NOTES

Why this path fits your profile

- Cross-domain pattern recognition — PQC sits at the intersection of math, CS, physics, and security policy. Very few people can hold all four simultaneously. Your top 3-4% cross-domain score means you can.
- Learning velocity — the math ramp (Phase 1-2) is where most people stall. Your demonstrated ability to go from zero to functional in novel domains (Forge3D in 4 hours) suggests you'll compress the timeline significantly.
- Tool mastery — your meta-tooling instinct (17 custom skills, shared infrastructure) maps directly to building the scanner. You already think in terms of reusable, self-improving systems.
- Solo builder profile — quantum security consulting doesn't require a team. One expert with a good tool and credentials can serve multiple enterprise clients.

Known risk from your profile

- Scope creep — you'll want to branch into quantum ML, quantum game dev, quantum everything. This roadmap is deliberately linear. Phase 1 before Phase 2. No skipping.
- Frustration tolerance (top 14%) — Phases 1-2 are heavy math grind with no flashy payoff. This is where the 90-day focus discipline gets tested.
- Follow-through — your polymath sprinter profile means you accelerate fast and then pivot to the next thing. This roadmap requires sustained depth on one domain for months. The deliverables at each phase are designed as checkpoints to keep momentum visible.

T3 AI Intersection

Phase 5 lattice math (LWE, Ring-LWE) uses the same vector space concepts as transformer attention mechanisms. Phase 3 quantum gates are unitary matrix operations — same linear algebra as neural network weight matrices. This roadmap doesn't compete with T3 AI; it reinforces it. The math is the same math.

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